ASSIGNMENT- 07

Bachelor of Computer Applications (Hons. AI & ML)
Semester- III

Subject: Database Management System

Topic: Sample Data Insertion

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1. Aim

To insert realistic sample data into all tables of the Paranormal Investigation Agency Management System using SQL `INSERT` statements while maintaining primary key, foreign key, and other integrity constraints.

2. Introduction

Sample data insertion is an important step in database implementation. It helps verify whether tables, relationships, and constraints work correctly. In this project, sample records are inserted into all 11 tables of the Paranormal Investigation Agency Management System.

The records represent clients, investigators, investigation locations, case categories, equipment, cases, assigned investigators, equipment usage, evidence, investigation reports, and payments.

3. SQL Queries for Sample Data Insertion

The following SQL commands use MySQL syntax. Execute them in the given order to ensure that referenced records exist before inserting dependent records.

3.1 Insert Data into Clients

INSERT INTO Clients

(client_id, name, email, phone, address, registration_date)

VALUES

(1, 'Arun Mathew', 'arun@example.com', '9876501001',

'Bengaluru, Karnataka', '2026-01-10'),

(2, 'Sneha Nair', 'sneha@example.com', '9876501002',

'Kochi, Kerala', '2026-01-15'),

(3, 'Rahul Menon', 'rahul@example.com', '9876501003',

'Mysuru, Karnataka', '2026-02-02'),

(4, 'Anjali Joseph', 'anjali@example.com', '9876501004',

'Thrissur, Kerala', '2026-02-18'),

(5, 'Vivek Rao', 'vivek@example.com', '9876501005',

'Chennai, Tamil Nadu', '2026-03-05');

Explanation: This query inserts five clients who have registered with the paranormal investigation agency.

3.2 Insert Data into Investigators

```
INSERT INTO Investigators

(investigation_id, name, speciliazation, phone, email,

 experience_level, date_joined)

VALUES

(1, 'Daniel Thomas', 'Paranormal Research', '9876511001',

 'daniel@example.com', 'Senior', '2023-06-10'),

(2, 'Meera Krishnan', 'Digital Evidence', '9876511002',

 'meera@example.com', 'Intermediate', '2024-02-15'),

(3, 'Joseph Varghese', 'Historical Investigations', '9876511003',

 'joseph@example.com', 'Senior', '2022-11-20'),

(4, 'Neha Sharma', 'Audio Analysis', '9876511004',

 'neha@example.com', 'Junior', '2025-01-08'),

(5, 'Adithya Kumar', 'Field Investigation', '9876511005',

 'adithya@example.com', 'Intermediate', '2024-08-12');
```

Explanation: This query stores investigator details, including their specializations and experience levels.

3.3 Insert Data into Locations

```
INSERT INTO Locations

(location_id, addess, location_type, year_built, known_history)

VALUES

(1, 'Old Heritage Mansion, Bengaluru', 'Residential', 1920,

 'Reported unexplained sounds'),
```

(2, 'Abandoned Factory, Kochi', 'Industrial', 1955,

'Reported unusual activity'),

(3, 'Historic Bungalow, Mysuru', 'Residential', 1890,

'Local stories of unexplained events'),

(4, 'Old Theatre, Thrissur', 'Commercial', 1940,

'Reported strange noises'),

(5, 'Heritage Building, Chennai', 'Historical', 1910,

'Historical reports of unusual occurrences');

Explanation: This query records the locations where investigations may take place.

3.4 Insert Data into Case_Categories

INSERT INTO Case_Categories

(Category_id, category_name, description)

VALUES

(1, 'Unexplained Sounds', 'Investigation of unusual sounds'),

(2, 'Apparition Reports', 'Investigation of reported apparitions'),

(3, 'Electronic Disturbances', 'Investigation of unusual electronic activity'),

(4, 'Historical Investigation', 'Research into historical incidents'),

(5, 'Environmental Anomalies', 'Investigation of unusual environmental readings');

Explanation: This query creates categories to classify different types of investigation cases.

3.5 Insert Data into Equipment

INSERT INTO Equipment

(equipment_id, equipment_name, equipment_type,

condition_status, purchase_date)

VALUES

(1, 'Digital Audio Recorder', 'Audio', 'Good', '2025-01-10'),

(2, 'Night Vision Camera', 'Camera', 'Good', '2025-02-15'),

(3, 'EMF Meter', 'Detection', 'Good', '2025-03-20'),

(4, 'Thermal Camera', 'Imaging', 'Fair', '2025-04-05'),

(5, 'Portable Microphone', 'Audio', 'Good', '2025-05-12');

Explanation: This query stores the equipment available for use during investigations.

3.6 Insert Data into Cases

INSERT INTO Cases

(Case_id, Client_id, Location_id, Category_id,

Date_reported, Investigation_date, Case_status, Urgency_level)

VALUES

(101, 1, 1, 1, '2026-03-01', '2026-03-10', 'Completed', 'High'),

(102, 2, 2, 2, '2026-03-05', '2026-03-15', 'In Progress', 'Medium'),

(103, 3, 3, 3, '2026-03-12', '2026-03-22', 'Open', 'Low'),

(104, 4, 4, 4, '2026-03-18', '2026-03-28', 'Completed', 'Medium'),

(105, 5, 5, 5, '2026-03-25', '2026-04-05', 'In Progress', 'High');

Explanation: Each case is linked to an existing client, location, and category through foreign keys.

3.7 Insert Data into Case_Investigator

INSERT INTO Case_Investigator

(Case_id, Investigator_id, Role_in_case)

VALUES

(101, 1, 'Lead Investigator'),

(101, 2, 'Evidence Analyst'),

(102, 3, 'Lead Investigator'),

(102, 4, 'Audio Analyst'),

(103, 5, 'Field Investigator'),

(104, 1, 'Lead Investigator'),

(104, 3, 'Historical Researcher'),

(105, 2, 'Evidence Analyst'),

(105, 5, 'Field Investigator');

Explanation: This junction table assigns investigators to cases and supports the many-to-many relationship between Cases and Investigators.

3.8 Insert Data into Case_Equipment

INSERT INTO Case_Equipment

(Case_id, Equipment_id, Checkout_time, Return_time)

VALUES

(101, 1, '2026-03-10 09:00:00', '2026-03-10 18:00:00'),

(101, 3, '2026-03-10 09:15:00', '2026-03-10 18:00:00'),

(102, 2, '2026-03-15 10:00:00', '2026-03-15 17:00:00'),

(103, 4, '2026-03-22 08:30:00', '2026-03-22 16:30:00'),

(104, 5, '2026-03-28 09:00:00', '2026-03-28 17:30:00'),

(105, 3, '2026-04-05 10:00:00', '2026-04-05 18:00:00');

Explanation: This table records which equipment was used for each case and when it was checked out and returned.

3.9 Insert Data into Evidence

INSERT INTO Evidence

(Evidence_id, Case_id, Recorded_by, Evidence_type,

Description, File_refernce, timestamp)

VALUES

(1, 101, 1, 'Audio',

'Unusual sound recorded during the investigation',

'CASE101_AUDIO01.wav', '2026-03-10 14:30:00'),

(2, 101, 2, 'Image',

'Photograph of the investigation location',

'CASE101_IMAGE01.jpg', '2026-03-10 15:00:00'),

(3, 102, 3, 'Video',

'Video recorded during the site inspection',

'CASE102_VIDEO01.mp4', '2026-03-15 13:15:00'),

(4, 103, 5, 'Measurement',

'Electronic readings recorded at the location',

'CASE103_DATA01.csv', '2026-03-22 11:45:00'),

(5, 104, 1, 'Audio',

'Audio recording collected from the old theatre',

'CASE104_AUDIO01.wav', '2026-03-28 12:30:00'),

(6, 105, 2, 'Image',

'Photograph documenting the investigation site',

'CASE105_IMAGE01.jpg', '2026-04-05 14:00:00');

Explanation: This query stores evidence details and links each record to a case and the investigator who recorded it.

3.10 Insert Data into Reports

INSERT INTO Reports

(Report_id, Case_id, Prepared_by, Final_verdict, Summary, Report_date)

VALUES

(1, 101, 1, 'Inconclusive',

'The available recordings were reviewed, but no definitive explanation was established.',

'2026-03-12'),

(2, 104, 3, 'Unverified',

'Historical records and collected observations were reviewed. Further investigation may be required.',

'2026-03-30');

Explanation: Reports are inserted for the completed cases. Each report is associated with a case and the investigator who prepared it.

3.11 Insert Data into Payments

INSERT INTO Payments

(Payment_id, Case_id, Amount, Payment_mode, Payment_status, Payment_date)

VALUES

(1, 101, 15000.00, 'UPI', 'Paid', '2026-03-12'),

(2, 102, 10000.00, 'Card', 'Pending', '2026-03-15'),

(3, 103, 8000.00, 'Cash', 'Paid', '2026-03-22'),

(4, 104, 12000.00, 'Bank Transfer', 'Paid', '2026-03-30'),

(5, 105, 18000.00, 'UPI', 'Pending', '2026-04-05');

Explanation: This query stores payment information for each case, including the amount, payment mode, and payment status.

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4. Verification of Inserted Data

The following queries can be executed to verify that records have been inserted successfully.

SELECT * FROM Clients;

SELECT * FROM Investigators;

SELECT * FROM Locations;

SELECT * FROM Case_Categories;

SELECT * FROM Equipment;

```
SELECT * FROM Cases;

SELECT * FROM Case_Investigator;

SELECT * FROM Case_Equipment;

SELECT * FROM Evidence;

SELECT * FROM Reports;

SELECT * FROM Payments;
```

To check the total number of records in each table:

```
SELECT 'Clients' AS Table_Name, COUNT(*) AS Total_Records

FROM Clients

UNION ALL

SELECT 'Investigators', COUNT(*) FROM Investigators

UNION ALL

SELECT 'Locations', COUNT(*) FROM Locations

UNION ALL

SELECT 'Case_Categories', COUNT(*) FROM Case_Categories

UNION ALL

SELECT 'Equipment', COUNT(*) FROM Equipment

UNION ALL

SELECT 'Cases', COUNT(*) FROM Cases

UNION ALL

SELECT 'Case_Investigator', COUNT(*) FROM Case_Investigator

UNION ALL

SELECT 'Case_Equipment', COUNT(*) FROM Case_Equipment

UNION ALL

SELECT 'Evidence', COUNT(*) FROM Evidence
```

```
UNION ALL

SELECT 'Reports', COUNT(*) FROM Reports

UNION ALL

SELECT 'Payments', COUNT(*) FROM Payments;
```

Expected result: If all statements execute successfully in a fresh database, the counts should be:

Table Name	Expected Records
Clients	5
Investigators	5
Locations	5
Case_Categories	5
Equipment	5
Cases	5
Case_Investigator	9
Case_Equipment	6
Evidence	6

Reports	2
Payments	5

5. Integrity and Relationship Validation

The inserted data follows these rules:

- **Primary key:** Each record has a unique identifier.
- **Foreign key:** Case records refer to existing clients, locations, and categories.
- **Relationship integrity:** Investigators and equipment are assigned only to existing cases.
- **Evidence integrity:** Every evidence record refers to an existing case and investigator.
- **Report integrity:** Each report refers to an existing case and its preparing investigator.
- **Payment integrity:** Each payment refers to an existing case.
- **Data consistency:** Dates, descriptions, and status values are entered in appropriate formats.

Note: Execute the queries only after creating the tables from Assignment 5. If you have already inserted records with the same primary keys, use a fresh database or remove the previous sample records before running the statements again. The status values must also match any `CHECK` constraints configured in Assignment 6.

6. Result

Sample data was inserted into all 11 tables of the Paranormal Investigation Agency Management System using SQL `INSERT` statements. The records demonstrate how primary keys, foreign keys, and relationships maintain consistency across the database.

7. Conclusion

This assignment demonstrates how to populate a relational database with realistic sample records. The inserted data supports testing and verification of the database structure, constraints, and relationships. It also provides a foundation for performing SQL queries, generating reports, and managing investigation cases efficiently.